

Lesson 1.13: Lab 2, Sheet Pan Nachos with a Label Discussion

Unit: Unit 1, Kitchen Skills and Smart Eating (Nutrition and Wellness) **Topic:** 1.3 Nutrients, MyPlate, and Labels **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes (Day 1 prep and label math; Day 2 the lab) **Room:** FACS lab with kitchen stations and an oven (a no-kitchen version is below)

Standards

- NYS module line: NYS N&W 1. Culinary Management b) Demonstrate safety and sanitation procedures when handling food and equipment
- NYS module line: NYS N&W 1. Culinary Management d) Prepare a simple recipe
- NYS module line: NYS N&W 1. Culinary Management g) Prepare and serve a simple meal
- NYS module line: NYS N&W 2. Nutrition Across the Lifespan a) Identify nutrient groups, common sources of nutrients, and nutrient requirements across the lifespan
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE theme line: CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services
- NYS FACS Learning Standard 1 (Intermediate), Key Idea: Students will use an understanding of the elements of good nutrition to plan appropriate diets for themselves and others. They will know and use the appropriate tools and technologies for safe and healthy food preparation.; Performance Indicator A: Students understand the relationships among diet, health, and physical activities; evaluate their own eating patterns; and use appropriate technology and resources to make food selections and prepare simple, nutritious meals.; Performance Indicator B: Students apply principles of food safety and sanitation.
- CDOS Standard 3a Universal Foundation Skills: Interpersonal Skills
- National FCS 3.0: 14.3.3 Show the ability to select, store, prepare, and serve food that is nourishing and looks good on the plate.
- National FCS 3.0: 14.2.4 Examine sources of food and nutrition information, food labels included, for what they say about health.

(Four module lines is one more than the usual three. Lines 1. b), d), and g) are the lab; line 2. a) is the label task that runs while the pan bakes. The theme note from Lesson 1.11 applies: FCL 4. has no “labels” line, so f) is the informed-purchasing line and is quoted verbatim.)

Health class coordination

Health class typically covers: “choose healthy snacks” and sometimes a sodium or sugar limit as a number on a slide. Health does not cook.

This lesson goes further: students cook a real snack in a real kitchen, read the labels of the actual ingredients they used, compute the sodium in the plate in front of them, and judge it with a number. The label math is the piece no other class in the building does with a hot pan on the table.

Enduring understanding and essential question

Big idea: You can cook something fast, safe, and good, and still know exactly what is in it.

Reading the label of what you cooked is the honest version of reading a label. **Essential**

question: Is a snack you made yourself automatically healthier than one from a package?

Objectives

By the end of this lesson, students will be able to:

1. (Low: understand) Given the station card and their role card, state their own role's tasks in order and the two rules for the oven step (only the teacher opens the oven; the pan lands on a trivet and nobody touches it for one minute).
2. (Mid: apply) Working in a team of about seven with the 40-minute flowchart, prepare and serve sheet pan nachos with a salsa bar, following safety and sanitation procedures, with every role completed and the station passing the end check.
3. (High: evaluate) Using the chip, cheese, bean, and corn labels, calculate the sodium in one serving of their own nachos, compare it to the daily value, and judge whether this is an “often” snack or a “sometimes” snack with the number as evidence.

Vocabulary

Tier 2 (general academic): flowchart, role, compute, judge Tier 3 (FACS): sheet pan, drain, layer, trivet, oven mitt, sodium, percent daily value. See the unit vocabulary list for Turkish, Portuguese, and Spanish.

Materials and setup

Per station (three stations of about seven students):

- 1 bag restaurant style tortilla chips, 13 ounces (a station of four or five gets half a bag)
- 1 cup (4 ounces) shredded Mexican blend cheese, pre-measured into a covered cup by the teacher or measured by the Prep Cook
- 1/2 can black beans (15.5 ounce can), can pre-opened by the teacher, drained and rinsed by the Prep Cook in a colander
- 1/3 can corn (15.25 ounce can), can pre-opened by the teacher, drained only

- 1 half sheet pan or quarter sheet pan, 1 sheet of heavy duty foil pre-cut to fit, 1 colander, 1 dry measuring cup (1 cup), 1 large spoon, 1 spatula, 2 oven mitts, 1 trivet, 7 paper plates, 7 forks, napkins, 1 sanitizing spray and 2 cloths
- Station Card, Role Cards, and Nacho Label Math handout: Handout 01.13 - Station Card Sheet Pan Nachos.md, Handout 01.13 - Role Cards Lab 2.md, Handout 01.13 - Nacho Label Math.md
- The four empty packages (chip bag, cheese bag, bean can, corn can) with their Nutrition Facts panels, for the label task (the handout also prints the numbers)

Shared salsa bar, on a separate counter, set up before class and kept cold until the oven step:

- 1 jar salsa, 16 ounces, with a serving spoon
- 1 tub sour cream, 16 ounces, with a serving spoon
- Optional, no knives required: 1 can sliced black olives (drained), 1 bag pre-shredded lettuce.

Teacher station: oven preheated to 400 degrees F before students arrive, timer, oven mitts, clipboard with the lab rubric.

- Slides: Slides 01.13 - Lab 2 Sheet Pan Nachos with a Label Discussion.md (outline; build the deck from it; Day 1 and Day 2 are labeled)
- Setup notes: pre-cut foil the day before; pre-open one bean can and one corn can per station so no student touches a sharp lid; set the salsa bar before class; confirm the oven is at 400 before the bell; put oven mitts and a trivet at every station. Zero-knife lab: nothing is cut. Allergy check the day before: dairy (cheese, sour cream), and check the chip bag for shared-line warnings (some are made on lines with milk or wheat). A student who cannot eat dairy gets a foil-divided corner of the pan with no cheese and skips the sour cream.

Pre-assessment

The Kitchen Safety Exam (Topic 1.1) is the gate; nobody cooks without a pass on file. The Lab 1 rubric scores (Lesson 1.8) tell you which teams were slow on cleanup; give those teams the strongest Sanitation Lead. The Lesson 1.11 Day 2 exit tickets and the Spot the Trick handouts tell you who can do percent daily value math; seat one strong label reader at each station as the Label Reader. The Label Reading Quiz comes after this lab, in the first 15 minutes of the next period (see Assessment 01.3 - Label Reading Quiz.md).

Do now and hook (Time: Day 1, 5 min; Day 2, 5 min)

Day 1, on the board: “You are making nachos for seven people in 40 minutes with no knives. Write the steps in order, as many as you can.” Debrief: take one step from each of four students and write them in a column. The hook: show the 40-minute flowchart on the slide. “Here is the real order. Tomorrow you cook it. Today you learn your job, and you find out how much salt is in it before you eat it.”

Day 2, the do now is the setup routine: wash hands, tie back hair, apron on, role card in hand, station check against the station card list (everything on the counter, nothing missing). The Head Chef reports “station ready” to the teacher. The hook is the oven: “It is at 400. It is mine. Your pans go in when I say and come out when I say.”

Procedure

Day 1: prep, roles, and the label task

Step	Teacher will	Students will	Time
1. Direct instruction: roles and the station card	Assign teams (about seven per station) and hand out role cards: Head Chef, Sous Chef, Prep Cook, Sanitation Lead, Label Reader, Runner, Timekeeper and Safety Officer. Walk the station card step by step on the slide, naming which role does each step. Say the two oven rules and have the class repeat them.	Read their role card. Highlight their steps on the station card. Say their role’s first task to a partner.	8 min
2. Guided discussion: the salsa bar plan	Show the salsa bar layout. Explain the cold rule from Lesson 1.2: salsa and sour cream stay in the refrigerator until the pans go in the oven, and the Runner brings them to the bar then. Agree on the traffic pattern: one station at the bar at a time, Runner serves, plates come to the bar, not the bar to the plates.	Runners draw the traffic pattern on the back of the station card. Everyone writes the cold rule under the salsa bar box.	4 min
3. Learning activity:	Hand out the four	Read four real	15 min

the Nacho Label Math	packages and the Nacho Label Math handout. Task, led by the Label Reader with the whole team computing: for each ingredient, find the serving size and the sodium per serving; multiply by the servings the station uses (chips 13, cheese 4, beans 1.75, corn 1); add the four to get the sodium in the whole pan; divide by 7 for one person's share; add 2 tablespoons of salsa and 2 of sour cream; compare to the daily value of 2,300 mg. Model the first line (chips) on the board. Circulate.	labels. Fill the table. Compute the pan total, the per-person share, and the percent of a day. Answer the judgment question: often or sometimes, with the number.	
4. Share and safety review	Take the per-person number from each station (they should land close to each other; about 680 mg with salsa and sour cream, about 30 percent of a day). Ask why the beans are the biggest surprise and what rinsing does (it removes roughly 40 percent of the sodium). Then the safety review: hot pan path from oven to trivet, mitts, no reaching over a pan, no student at the	Reporters give the number. Class votes often or sometimes. Timekeepers read the two oven rules aloud.	4 min

	oven.		
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Day 2: the lab

Step	Teacher will	Students will	Time
1. Line the pan and layer the chips (minutes 5 to 10)	Start the timer. Call “line and layer.” Check each pan for even chips (one layer, not a mountain; the cheese has to reach the bottom).	Sous Chef lines the pan with foil and spreads the chips in an even layer. Runner brings the chips and cheese from the supply table. Prep Cook is at the sink.	5 min
2. Drain and top (minutes 10 to 16)	Watch the sink: beans drained and rinsed in the colander, corn drained only. Check each pan before it moves: beans and corn spread, cheese over everything.	Prep Cook drains and rinses beans, drains corn. Sous Chef spreads beans and corn, then cheese. Head Chef checks the pan against the station card and calls “pan ready.” Sanitation Lead wipes the counter and clears the cans.	6 min
3. Oven, 400 degrees F, 8 to 10 minutes (minutes 16 to 26)	Load all pans with mitts. Set the timer for 8 minutes and check at 8; pull at 10 at the latest. During the bake, run the label discussion at the front: hold up the chip bag and the bean can, ask each station for its per-person sodium number from Day 1, ask which one ingredient they would swap to cut it most, and what that would do to the	Runner and one other bring salsa and sour cream from the refrigerator to the bar. Everyone else sits at home base for the label discussion. Label Reader speaks for the station. Timekeeper watches the clock and calls “two minutes.”	10 min

	taste.		
4. Remove and plate (minutes 26 to 32)	Pull each pan with mitts and set it on that station's trivet. Say "one minute, hands off." Then release stations to plate: spatula onto plates, one station at a time to the salsa bar.	Head Chef portions with the spatula, about one seventh per plate. Runner takes the station's plates to the bar and serves salsa and sour cream (2 tablespoons each is a serving). Sanitation Lead starts the wash water.	6 min
5. Eat (minutes 32 to 36)	Sit with a plate. Ask the essential question at each station: is this healthier than the bag it came from? Score the rubric on the clipboard as you go (Objective 2 is scored here).	Eat. Answer the essential question at the table with the sodium number.	4 min

Questions to ask

Monitor understanding:

- What is your role's first step? What is your last one?
- Which ingredient has the most sodium per serving? Which has the most in the whole pan? Why are those different?
- Who opens the oven? Where does the pan go when it comes out?

Deepen learning:

- The beans are the healthiest thing on the pan and the saltiest. How can both be true?
- If you swapped the chips for a low sodium chip, how much would the per-person number drop? Is it worth it?
- Is a snack you made yourself automatically healthier than the bag? What decides that?

Check for understanding

Day 1, clipboard check during Step 3: for each station, yes or no on "reads serving size before sodium," "multiplied by the station's servings, not by servings per container," and "per-person number is between 600 and 750 mg." Day 2, the rubric on the clipboard: for

each student, the role card tasks done, the two oven rules kept, and the station check at the end. A station that finishes the wash before the bell with the counter wiped and the pan clean passes the station check.

Closure (Time: Day 1, 4 min; Day 2, 4 min)

Day 1: exit line. Each student writes one line: “My role is . **My first step is** . The per-person sodium number is ___ mg.” Three read aloud. Sort by role: any role with two or more blank first steps gets a reminder on the Day 2 slide.

Day 2: clean up and station check (minutes 36 to 40). Sanitation Lead runs the list on the station card: pan washed and dried, foil in the trash, counter sprayed and wiped, sink empty, mitts and trivet back, chairs in. Head Chef reports “station clean” to the teacher, who signs the station card. Exit line on the way out, spoken to the teacher: “Often or sometimes, and why?”

Differentiation and supports

- ELL: role cards have the tasks as numbered short lines with a picture in words for each; the station card has the same steps in Turkish, Portuguese, and Spanish on the back (see Vocabulary 01.md); the oven rules are two pictures on the slide; pair each ELL student with a partner in a linked role (Sous Chef with Prep Cook).
- IEP and 504 (general): a student who cannot stand for the whole period is the Label Reader or the Timekeeper at home base; the label math handout has the multiplication written as a fill-in with the servings already in the box; a calculator at every station; a student with a sensory issue around heat stays away from the trivet and plates last.
- Grade 6 support: the Nacho Label Math uses rounded numbers (beans as 2 servings, corn as 1) and asks for the pan total and the per-person share only; the percent of a day is given.
- Grade 8 stretch: compute the pan total for calories and saturated fat too, and rewrite the recipe with one swap (low sodium beans, reduced fat cheese, or half the chips) with the new per-person number; on Day 2, the Head Chef at a grade 8 station also plates one “MyPlate nacho plate” with a side of the salsa bar lettuce and reports whether it moved the plate closer to MyPlate.
- UDL checkpoint used: multiple means of action and expression (seven roles so every student has a physical task that fits them) and multiple means of representation (the flowchart as a slide, a station card, and a timer the teacher calls aloud).
- No-kitchen or no-machine alternative: Day 1 runs as written. Day 2 becomes “build a nacho plate”: each team gets the Nutrient Sort Cards plus six nacho ingredient cards (chips, cheese, beans, corn, salsa, sour cream) with the label numbers on the back. Teams build the plate, compute the sodium for the plate exactly as on Day 1, then rebuild it with two swaps to bring the sodium under 400 mg per person while keeping at least three MyPlate groups on the plate. The Sanitation Lead’s job becomes the recorder. If a microwave is available, a single microwave version works: chips, beans, corn, and cheese on a paper plate for 45 seconds, one plate per team, teacher at the microwave.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. State own role tasks and the two oven rules	N&W 1. b); HSW 4. c); Std 1 PI B	Role cards and station card, Day 1 Step 1; exit line	Formative; exit line checked; summative on the Lab 2 rubric, row 1
2. Prepare and serve nachos safely in a team, station passes the end check	N&W 1. b), d), g); HSW 4. c), e); Std 1 PI B; CDOS 3a Interpersonal Skills; National FCS 14.3.3	The lab, Day 2 Steps 1 to 5 and closure	Summative; Lab 2 rubric, row 2, scored on the clipboard during the lab
3. Compute the sodium in one serving and judge the snack with evidence	N&W 2. a); FCL 4. f); Std 1 PI A; National FCS 14.2.4	Nacho Label Math, Day 1 Step 3; label discussion, Day 2 Step 3	Summative; Lab 2 rubric, row 3, scored from the handout and the Day 2 discussion

Lab 2 rubric (the lab rubric pattern from the grading plan: three objectives plus a safety and sanitation row, each 1 to 4, 16 points; counts in Labs and Projects)

Criterion	Not Yet 1	Approaching 2	Meets 3	Exceeds 4
1. Role and oven rules	Could not state own role's steps or an oven rule; needed the card read aloud	Stated the role's steps out of order or one oven rule	Stated own role's steps in order and both oven rules	Stated own role and a teammate's role, both oven rules, and why each rule exists
2. Safe teamwork and a clean station	Skipped role tasks, or a safety or sanitation step was missed (hands, hair, hot pan, sink), or the station failed the end check	Did most role tasks; one reminder on safety or sanitation; station passed the check late	Did every role task, kept every safety and sanitation step, station passed the end check on time	Did every role task, helped another role without being asked, station passed the check early with the counter and sink spotless
3. Label math and judgment	Table incomplete or the per-person number missing	Table complete but one computation wrong, or the judgment has	Table complete, per-person number within range, judgment states	All of Meets plus a swap that cuts the sodium, with the new

		no number	often or sometimes with the number	number computed
4. Safety and sanitation (every lab, same row all year)	A safety violation: hands not washed, hair loose over food, bare hand on a hot pan, a student at the oven, food dropped and served, or a station left dirty after the check	Hands, hair, and apron done, but one reminder needed during the lab (a towel near the oven, reaching over a pan, a wet floor left)	Washed hands at the start; hair and apron done; mitts and trivet used for every hot pan; station passed the check the first time	Meets, and called out a hazard for the team before the teacher saw it, or fixed one at another station without being asked
Total				/16

How to use: score rows 2 and 4 on the clipboard during Day 2; score rows 1 and 3 from the Day 1 exit line and the handout after class. Write one line in the notes box for the next lab (what this student should try next). A safety violation is a 1 on row 4 and a zero for that lab's participation, per the grading plan, and a conversation.

Homework

None. Optional: read the label on one snack at home and compute the sodium in your usual portion, not the serving. Bring the number.

Connections

Inside the building: the math teacher (the same ratio work as Lesson 1.11, now with a pan on the table) and the cafeteria manager, who can show the sodium number on the school lunch menu if the district posts one. FACS gives the cafeteria a class that will read the posted numbers; the cafeteria gives FACS a real menu for the \$40 project comparison in Topic 1.4.

Sources

- NYS Middle Level CTE FACS Content Module 05, Nutrition and Wellness (nycetecenter.org, June 2018), lines 1. b), 1. d), 1. g), and 2. a). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 04, Health, Safety, and Wellness, lines 4. c) and 4. e); Theme Module 03, Financial and Consumer Literacy, line 4. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 1, Intermediate, FACS strand. Wording per 02 Standards/STANDARDS SOURCES - web research.md, section 2.
- CDOS Standard 3a, Interpersonal Skills. Same file.

- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 14.3.3 and 14.2.4.
- Sodium daily value (2,300 mg) and the 5 and 20 rule: U.S. Food and Drug Administration, “How to Understand and Use the Nutrition Facts Label” ([fda.gov](https://www.fda.gov/food/understand-use-nutrition-facts-label)). Rinsing canned beans removes about 41 percent of the sodium: Duyff, Jones, and Mount, “Sodium reduction in canned beans after draining, rinsing,” *Journal of Culinary Science and Technology*, 2011.
- My own materials: my Sheet Pan Nachos with Salsa Bar teacher sheet (the 40-minute flowchart, ingredient amounts per station, prep notes, the zero-knife policy, and the “Nutrition Facts discussion while it bakes” idea). Rewritten here for three stations of seven with the teacher on oven duty. The label numbers on the handouts are typical values for store brand products and must be replaced with the numbers on the packages you actually buy.

Teacher notes

- If Day 1 runs short, cut Step 2 (the salsa bar plan) to one minute and let the Runners figure out the traffic pattern on Day 2 with you at the bar. Never cut Step 3; the label math is Objective 3 and the rubric row.
- If Day 2 runs short, the bake is fixed at 8 to 10 minutes; the only place to save time is plating. Have the Runners pre-set plates at the bar during the bake.
- Per-station cost, estimated at Long Island supermarket store brand prices, September 2026: chips 13 oz \$3.99; cheese 8 oz \$3.49 (half used; buy one bag for two stations, \$1.75); black beans \$1.29 (half used; one can for two stations, \$0.65); corn \$1.19; salsa 16 oz \$3.29 shared by three stations, \$1.10; sour cream 16 oz \$2.49 shared, \$0.83; foil, about \$0.30. About \$9.80 per station buying shared items across the class, or about \$12.20 if every station gets its own cheese bag and bean can. Optional olives and lettuce add about \$1.35 per station. A class of three stations runs about \$30 to \$41, well under the \$20 per station line.
- Common mistake: a pan with the chips piled in a mound, so the middle stays cold and the cheese never reaches the bottom. Say “one layer, you should see foil between some chips” at Step 1.
- Second common mistake: the Prep Cook rinses the corn too and it goes watery, or does not rinse the beans and the pan tastes of the can. Beans: drain and rinse. Corn: drain only. It is on the station card twice.
- The teacher at the oven is not a suggestion in this version. My original sheet had students rotate mitts and pull their own pans; with three stations of seven and grade 6 in the room, the teacher pulls.
- The label discussion during the bake is the part that makes this a nutrition lesson and not just a snack. Do not skip it to “let them relax.” Ten minutes is enough for the number, the swap question, and the beans surprise.
- Keep the four empty packages. They are the Label Reading Quiz review set and one of the stations in Lesson 1.16 (unit price).